Semi-Automating the Passivation of Microneedle Arrays: Controlled Developer Immersion vs. Manual Application

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Background

- Microneedle arrays (MNAs) are small needle-like structures that painlessly pierce the outer skin to deliver therapeutics or sense biomarkers beneath it [2]. Needles are tens to hundreds of micrometres tall, arranged in arrays spaced 300–500 µm apart (see **Figure 1** for a schematic).
- To function as a sensor, the base of each microneedle is insulated with a passivation layer, while the shaft and tip remain exposed for a sensing species to be deposited [3]. The exposed region defines the active sensing area [1], [3].
- As exposed tip height sets the active sensing area, it is critical that it is uniform across all microneedles in the array for the sensor to respond consistently.
- Applying the passivation layer by hand is time consuming and difficult to control, producing unreliable and poorly repeatable results.

Objectives

1. Automate the passivation process for microneedle arrays.
2. Measure exposed tip height and compare the automated and manual processes.
3. Identify the process variables that limit consistency of exposed tip height.
4. Produce documented calibration and operation procedures for the system.

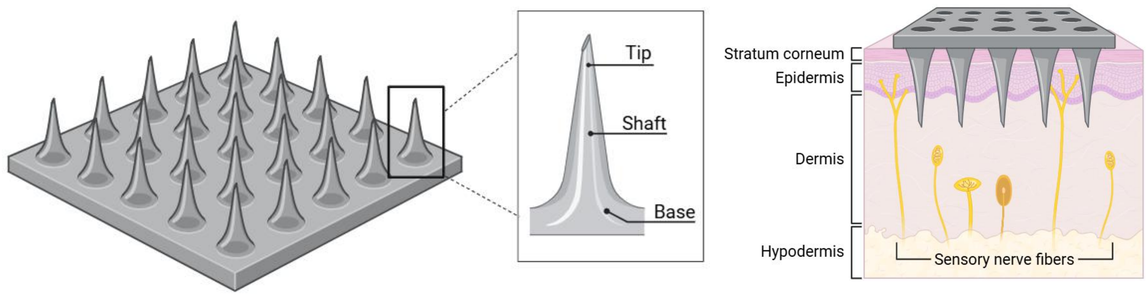


Figure 1: Labeled schematic of microneedle array (left), labeled schematic of microneedle array in skin (right)

Methods & Design Evolution

Initial approach

The system was built on a modified 3D printer frame with a three-axis stage resolving 25 µm horizontally and 5 µm vertically. It was first designed to dispense SU-8 **(a photocurable biocompatible polymer)** droplet-by-droplet through a pneumatic solenoid valve, which could not deliver the required volume control:

- **Solenoid valve response floor** — minimum reliable pulse ~100 ms; a hardware limit of the valve, not of the control software
- **No backpressure control** — fluid continued to flow after the valve closed and the meniscus did not reset at the tip
- **Viscosity did not resolve it** — SU-8 2000.5, 2002, and 2005 were each tested; none produced discrete droplets at the required scale
- **Surface forces dominate** — capillary forces between adjacent needles distributed SU-8 laterally regardless of the volume delivered

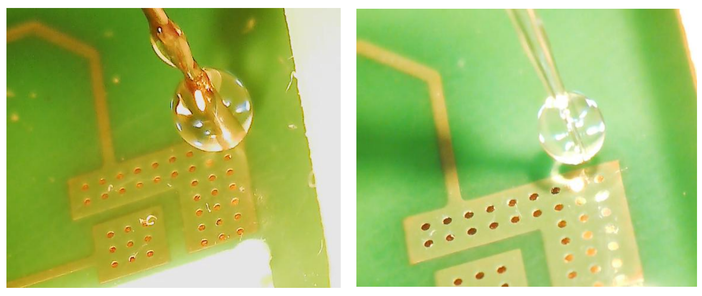


Figure 2: Droplet formation on a 30G needle (left) and droplet formation on a microcapillary (right). Minimum deliverable volume greatly exceeds the needle pitch

Manual process

Performed entirely by hand in a clean room:

1. Apply SU-8 with a micropipette; allow to wick between needles
2. Soft bake at 110° C to evaporate solvent in SU-8
3. Apply a small drop of developer to clear SU-8 from the tips
4. Rinse in isopropyl alcohol (IPA)
5. UV cure and hard bake at 130° C

The develop step sets final exposed tip height. Applied by hand, its volume, contact area, and dwell time vary between arrays.

Automated process (semi-automatic)

The same three-axis stage was retained. SU-8 application remains manual; the develop and rinse are automated:

1. Apply SU-8 with a micropipette; allow to wick between needles
2. Soft bake at 110° C to evaporate solvent in SU-8
3. **Automated:** array held face-down on a vacuum chuck, lowered into a fixed-dose developer well to a recorded depth and dwell time
4. **Automated:** transits to a second station; dipped in an IPA bath to rinse
5. UV cure and hard bake at 130° C

Controlled immersion fixes immersion depth, dwell time, and wetted area, all set in the software.

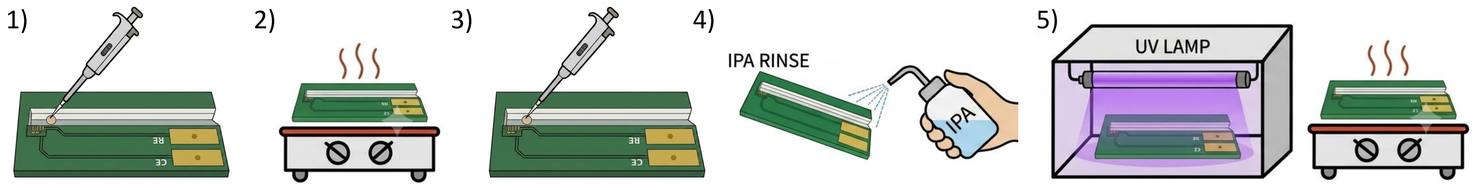


Figure 3: Step by step process of the manual process

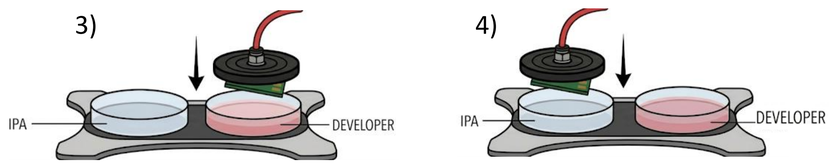


Figure 4: Automated process steps 3) and 4)

Assessment: exposed tip height measured by desktop scanning electron microscope (SEM, Thermo Scientific Phenom XL G2) on arrays passivated by both processes.

Results

- Positioning hardware assembled and calibrated; commanded travel verified against measured travel on all three axes
- Calibration and operation procedures documented for system handoff

Exposed Tip Height by Process

Method	Mean	S.D.	Range	n
Automated	242.5	3.3	238.9 - 245.4	3
Manual	249.0	3.7	246.8 - 253.3	3

All values in µm. n = number of arrays, 8 needles measured per array

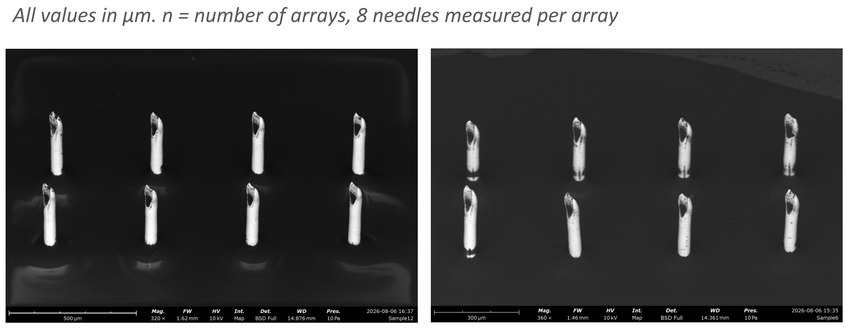


Figure 5: SEM images of automated process (left) and manual process (right)

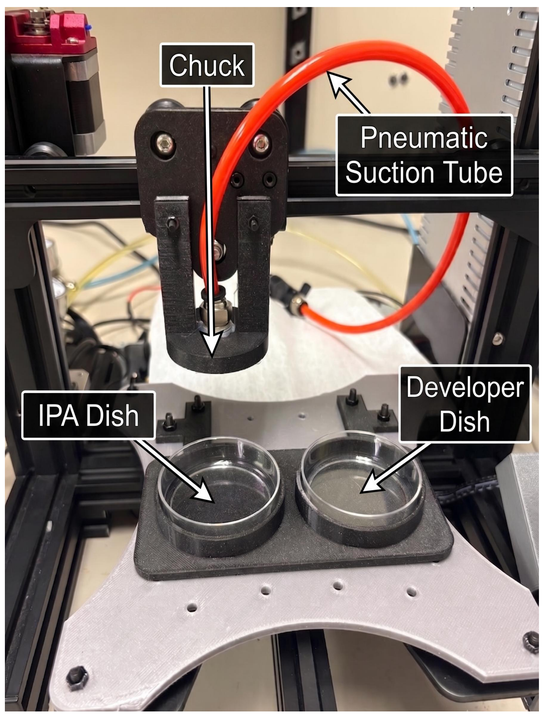


Figure 6: Labeled full system

Discussion & Future Work

Automated immersion produced spread comparable to manual application (S.D. 3.3 µm vs. 3.7 µm across n = 3); the two are not distinguishable at this sample size. Residual variability in the automated process is attributed primarily to fixture planarity: the vacuum chuck is not level across its face, so immersion depth and the extent of development vary across the array.

Reservoir surface height and dwell-time repeatability are additional contributors. These are correctable in hardware and do not reflect a limitation of controlled immersion itself.

More broadly, development, not dispensing, is the controlling step in setting exposed tip height on densely spaced arrays. A repeatable, documented passivation procedure reduces variability early in device fabrication and supports more reproducible microneedle biosensor research.

Future work

- Correct vacuum chuck planarity and verify immersion depth across the array
- Further calibrate dwell time and immersion depth to tune absolute exposed tip height
- Expand validation to more arrays and needle geometries

References

[1] K. Haider, T. Tilly, V. Coyle, and C. Dalton, “Scalable and customizable trimaterial microneedle electrochemical biosensor platform with cleanroom-free fabrication,” IEEE Sensors Letters, vol. 9, no. 9, pp. 1–4, Sep. 2025.
[2] K. Haider and C. Dalton, “Recent developments in microneedle biosensors for biomedical and agricultural applications,” Micromachines, vol. 16, no. 8, Art. no. 929, Aug. 2025.
[3] The Dalton Lab – Microneedle Research Group, “Automated photopolymer micro-dispenser: Technical design report,” University of Calgary, Calgary, AB, Canada, 2026.

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